

Exam Date & Time: 16-Nov-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Introduction To Extended Reality Application Development and Digital Twins [IT_4325]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Explain the concept of Extended Reality and summarize how it integrates with Digital Twin technology. (3)
 - A)
 - B) Differentiate among the four types of modelling used in XR and illustrate their relevance in rendering. (3)
 - C) Scenario: You are part of a startup building a remote-training simulator using VR headsets. Analyze the essential hardware components required to achieve realistic immersion and justify your selection. (4)
- 2) Explain the concept of embodiment and discuss how it affects user presence in the Metaverse. (4)
 - A)
 - B) Illustrate how distributed VR architectures enable interaction between multiple users. (3)
 - C) Scenario: You are planning a VR training platform for remote teams. Analyze whether a client-server or peer-to-peer architecture would be more efficient and justify your answer. (3)
- 3) Explain multimodal interaction in XR and identify advantages of gesture-based interfaces. (3)
 - A)
 - B) Discuss how graphics, sound, and haptic feedback displays enhance user immersion. (3)
 - C) Scenario: In a VR rehabilitation system, users interact with virtual objects using hand gestures. Analyze how navigation and manipulation interfaces can be designed to maximize comfort and minimize fatigue. (4)

- 4) Describe multiple-camera tracking and discuss its application in full-body motion capture. (4)
- A)
- B) Explain how IMU and lighthouse systems improve spatial tracking accuracy. (3)
- C) Scenario: In an outdoor AR exploration app, evaluate how GPS-based tracking compares with SLAM for position accuracy. (3)
- 5) Explain the role of a game engine in designing virtual worlds and summarize its key components. (3)
- A)
- B) Demonstrate how scripting in Unity (C#) and Unreal (Blueprint) enhances interactivity in XR applications. (3)
- C) Scenario: You are leading a team developing a mixed-reality educational game. Design a brief approach outlining how physics simulation and animation contribute to an engaging learning experience. (4)

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Event Name	Subject
058-END SEMSTER-ODD-2025	IT_4325:Introduction To Extended Reality Application Development and Digital Twins

#	Course Outcome	Description	Marks
1	1	CO 1	11
2	2	CO 2	9
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* NA - CO Not Selected

Event Name	Subject
058-END SEMSTER-ODD-2025	IT_4325:Introduction To Extended Reality Application Development and Digital Twins

#	Level	No of Questions	Marks (%)
1	Remember	1	4(8%)
2	Understand	3	10(20%)
3	Application	7	22(44%)

4	Analysis	4	14(28%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)